

WORKED EXAMPLE

A score from two input features

Use the exact example in Lecture 2, PDF page 13.

$$z = \theta_0 + \theta_1 x_1 + \theta_2 x_2 = -3 + x_1 + x_2$$

$$p = h_{\theta}(x) = g(z) = \frac{1}{1 + e^{-z}}$$

The lecture calls the probability h and the sigmoid g . We write the same probability as p . e is approximately 2.718.

$$(\theta_0, \theta_1, \theta_2) = (-3, 1, 1), \quad x_0 = 1$$

$x_0 = 1$ supplies the intercept. x_1 and x_2 are the two measured features.

These parameters are fixed for illustration. We are making predictions, not fitting the parameters in this lesson.

WORKED EXAMPLE

Calculate scores, probabilities and labels

Lecture 2, page 12: predict class 1 when p is at least 0.5.

$$(1, 1): z = -3 + 1 + 1 = -1, \quad p = \frac{1}{1 + e^1} \approx 0.268941 \Rightarrow 0$$

$$(1, 2): z = -3 + 1 + 2 = 0, \quad p = \frac{1}{1 + e^0} = 0.5 \Rightarrow 1$$

$$(2, 2): z = -3 + 2 + 2 = 1, \quad p = \frac{1}{1 + e^{-1}} \approx 0.731059 \Rightarrow 1$$

The arrow at the end gives the predicted class label. The decimal before it is a probability, not a class label.

A score of zero is a tie: our stated convention assigns the boundary to class 1.

WORKED EXAMPLE

Find the boundary without guessing

The decision boundary is the set of inputs where $p = 0.5$, equivalently $z = 0$.

$$-3 + x_1 + x_2 = 0 \implies x_2 = 3 - x_1$$

$$x_1 = 0 \Rightarrow x_2 = 3, \quad x_2 = 0 \Rightarrow x_1 = 3$$

Plot (0, 3) and (3, 0), then draw the straight line through them.

$$x_1 + x_2 \geq 3 \Rightarrow z \geq 0 \Rightarrow p \geq 0.5 \Rightarrow \hat{y} = 1$$

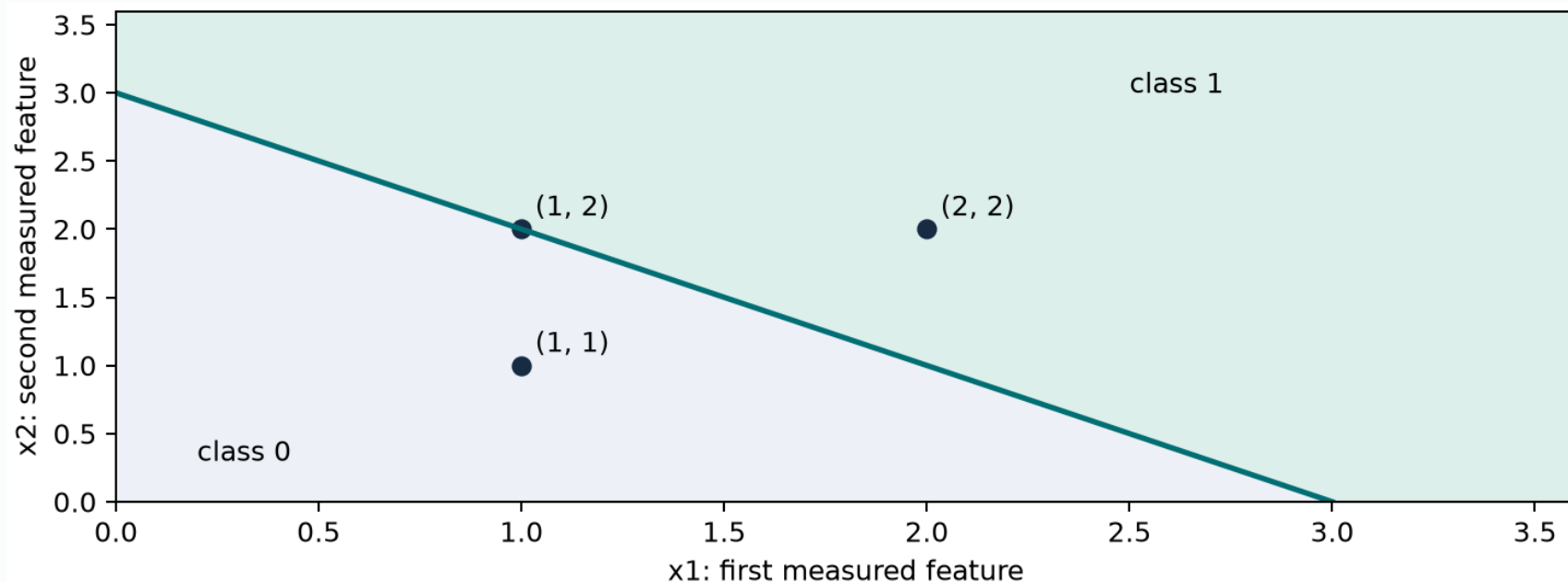
$$x_1 + x_2 < 3 \Rightarrow z < 0 \Rightarrow p < 0.5 \Rightarrow \hat{y} = 0$$

The boundary is a line in the input plane. It is not the S-shaped sigmoid graph.

WORKED EXAMPLE

Read the two sides of the line

The shaded regions are predicted classes; the dots are our three test inputs.



Moving across the line changes the predicted label. Probability varies smoothly even though the thresholded label changes abruptly.

WORKED EXAMPLE

Compute new features from the same inputs

Lecture 2, page 14 adds squared features. No new measured input is needed.

$$z = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2$$

$$(\theta_0, \theta_1, \theta_2, \theta_3, \theta_4) = (-1, 0, 0, 1, 1)$$

$$z = -1 + 0x_1 + 0x_2 + 1x_1^2 + 1x_2^2 = -1 + x_1^2 + x_2^2$$

The transformed feature list is (1, x_1 , x_2 , x_1 squared, x_2 squared). Square each measured input before multiplying its coefficient.

The score is still linear in its parameters: each theta multiplies a known feature. Its boundary can be curved in the original input plane.

WORKED EXAMPLE

Predict inside, on and outside the circle

Use the same sigmoid and the same probability threshold.

$$(0, 0): z = -1 + 0^2 + 0^2 = -1, \quad p = \frac{1}{1 + e^1} \approx 0.268941 \Rightarrow 0$$

$$(1, 0): z = -1 + 1^2 + 0^2 = 0, \quad p = \frac{1}{1 + e^0} = 0.5 \Rightarrow 1$$

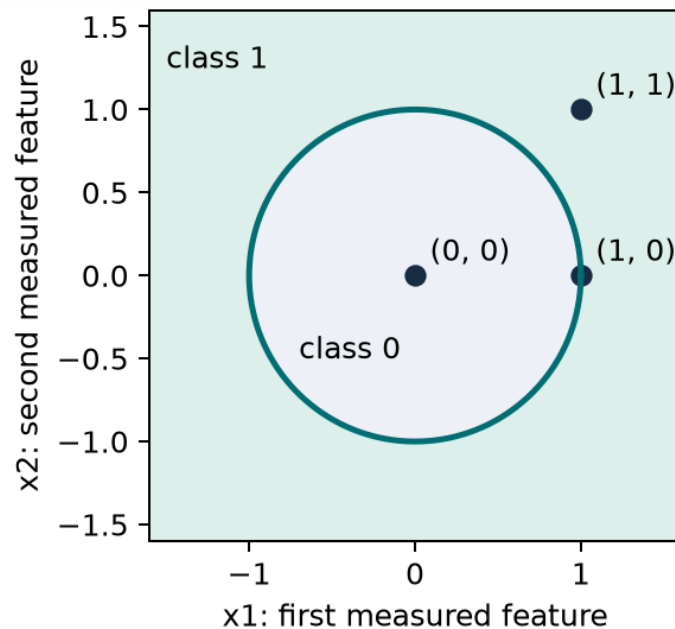
$$(1, 1): z = -1 + 1^2 + 1^2 = 1, \quad p = \frac{1}{1 + e^{-1}} \approx 0.731059 \Rightarrow 1$$

Only the score formula changed. A negative score still means $p < 0.5$; a positive score still means $p > 0.5$.

WORKED EXAMPLE

A circular decision boundary

Set $z = 0$: $x_1^2 + x_2^2 = 1$, a circle of radius 1.



Inside: $x_1^2 + x_2^2 < 1$ gives class 0. Outside and on the circle: at least 1 gives class 1.

INDEPENDENT PRACTICE

Your turn: compare the two models

Use the threshold $p \geq 0.5$ for class 1 throughout.

$$(x_1, x_2) = (0.5, 0.5)$$

1. For $z = -3 + x_1 + x_2$, calculate score, probability and class.
2. For $z = -1 + x_1^2 + x_2^2$, repeat the calculation.
3. In the curved model, change only the intercept from -1 to -4. Derive the new boundary and its radius.
4. What class do points exactly on that new boundary receive?

Keep probability values when calculating a loss later; do not replace them by 0 or 1.

WORKED ANSWER

Practice answer: scores and geometry

Changing a feature transformation changes the shapes the model can represent.

$$z_{\text{line}} = -3 + 0.5 + 0.5 = -2, \quad p = \frac{1}{1 + e^2} \approx 0.119203 \Rightarrow 0$$

$$z_{\text{circle}} = -1 + (0.5)^2 + (0.5)^2 = -0.5$$

$$p = \frac{1}{1 + e^{0.5}} \approx 0.377541 \Rightarrow 0$$

$$-4 + x_1^2 + x_2^2 = 0 \Rightarrow x_1^2 + x_2^2 = 4 \Rightarrow R = \sqrt{4} = 2$$

On the new circle, $z = 0$ and $p = 0.5$, so the label is 1 by our convention.

Avoid: forgetting the intercept; adding features before squaring them; confusing a probability with a label.